AHMED FERGANEY

AI & Data Engineer (LLMs, RAG, Backend Systems)

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PROFILE

AI Engineer specializing in LLM systems, RAG pipelines, and distributed AI backends. Built production-grade systems with sub-150ms latency and multi-agent orchestration workflows. Experienced in bridging data engineering, backend systems, and applied AI to deliver scalable, real-world solutions.

EDUCATION

B.Sc. in Mechanical Engineering | Banha University | 2016 – 2022

- Academic Standing: Ranked 5th out of 120 in the Mechanical Engineering Department

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, Deep Learning, NLP, LLM Engineering, AI Agents, Prompt Engineering, Fine-tuning, RAG, LangChain, Model Deployment, Transfer Learning, Transformer Architectures, Hyperparameter Optimization, PyTorch, TensorFlow

Data Analysis & BI: DAX, Power Query, ETL Development, SSIS, SSAS, SSRS, Data Warehousing, Data Cleaning, Data Visualization, Trend & Pattern Analysis, KPI Reporting.

Mathematics & Statistics: Statistics, Linear Algebra, Numerical Optimization

Databases & Data Engineering: Microsoft SQL Server, PostgreSQL, MongoDB, PGvector, Query Optimization, Relational & NoSQL Design, Database Design

Backend & Distributed Systems: FastAPI, RESTful API Design, Microservices Architecture, SQLAlchemy ORM, Pydantic Data Validation, Message Brokers (RabbitMQ), Celery, Redis

DevOps & Cloud: Docker, Kubernetes, Cloud Deployment, CI/CD Automation, Observability (Prometheus, Grafana)

Programming & Foundations: Python, C/C++, SQL, Bash, OOP, Data Structures & Algorithms

Tools: Power BI, MS Office, Postman, Git/GitHub, Jira, Trello

CERTIFICATIONS

- AI Engineer Core Track (LLM Engineering, RAG, QLoRA, AI Agents) — Udemy (2026, in progress)
- Prompt Engineering & RAG — Eng. Abu Bakr Soliman (2026)
- Databases, CCNA, Cloud Computing, Kubernetes, Jenkins & Redis — Udemy (2025)
- Fundamentals of Backend Engineering — Udemy (2025)
- Computer Vision with Python, TensorFlow & PyTorch — freeCodeCamp (2025)
- CNNs, RNNs, LSTMs, GRUs, NLP & Transformers — Stanford University (2024)
- Machine Learning & Deep Learning Diploma — Eng. Hesham Asem (2024)
- FastAPI, Data Structures, Algorithms & OOP — freeCodeCamp (2024)
- Statistics, Linear Algebra & Numerical Optimization — Dr. Hatem Elattar (2024)
- Embedded Linux, CMake, GDB, QEMU — Eng. Moatasem Elsayed (2024)
- Embedded Systems Diploma — Eng. Ahmed Abd El-Ghafar (2023)

PROFESSIONAL EXPERIENCE

Product Development Engineer | Kandil Glass | 10th of Ramadan City, Egypt *06/2026 – Present*

- Built end-to-end data analytics solutions by modeling production data into a star-schema warehouse, developed Power BI dashboards with DAX to track KPIs, uncover trends, and deliver actionable insights for data-driven decision-making.

Operational Excellence Engineer | Midea Electric | 6th of October City, Egypt *01/2025 – 05/2026*

- Analyzed operational data and built Power BI dashboards to track KPIs, downtime, and productivity trends.

Process Quality Engineer | Methode Electronics | Nasr City, Egypt *12/2023 – 12/2024*

- Collected and analyzed quality data to identify defect trends, building Excel reports for cross-functional teams.

Continuous improvement Engineer | Jac | Badr City, Egypt *03/2023 – 11/2023*

- Applied data-driven analysis of production KPIs and process metrics to optimize manufacturing performance, improve OEE, and reduce defects.

PROJECTS

Fine-Tuning Large Language Models for Arabic News Processing [GitHub]

- Fine-tuned Qwen2.5-1.5B-Instruct model on 2,700+ samples using LoRA technique with LLaMA-Factory, implementing structured entity extraction and translation tasks for Arabic news processing
- Deployed production-grade inference server using vLLM with LoRA adapter integration, achieving 80% GPU utilization and optimized throughput for real-time text processing applications

Multi-Agent LLM Orchestration System (LangGraph) [GitHub]

- Designed and implemented a multi-agent LLM pipeline (Planner, Researcher, Writer, Critic, Finalizer) using LangGraph, enabling iterative reasoning with structured state management and conditional feedback loops.

Async Document Intelligence RAG System [GitHub]

- Designed a distributed AI backend using FastAPI, Celery, and RabbitMQ for asynchronous document ingestion and query execution
- Built a retrieval + reasoning pipeline integrating LLMs with vector search (Qdrant/pgvector) for multi-step question answering
- Implemented async task queues (Celery workers) for scalable AI workloads, decoupling ingestion, embedding, and inference services across distributed workers
- Optimized API latency to <150ms (8.6x improvement over synchronous blocking) using FastAPI async endpoints, non-blocking I/O, Redis caching, and connection pooling
- Integrated observability with Prometheus + Grafana for real-time monitoring of task queues, vector DB performance, and LLM latency

Data Warehouse & BI Reporting — AdventureWorks (SSIS, Power BI) [GitHub]

- Designed a Kimball star-schema data warehouse (FactSales + 4 dimensions) from the AdventureWorks OLTP database.
- Built a full SSIS ETL package (Control Flow + Data Flow) to extract, transform, and load data; deployed to the SSIS Catalog and scheduled daily via SQL Server Agent.
- Developed Power BI dashboards with DAX measures (Revenue, YTD, YoY growth) covering sales trends, product performance, and territory analysis.

Transfer Learning for Image Classification [GitHub]

- Implemented transfer learning pipelines using pre-trained models (ResNet, VGG, EfficientNet) with fine-tuning strategies, data augmentation, and hyperparameter optimization to achieve production-ready classification accuracy

Bilingual SMS Spam Classification using NLP and Machine Learning [GitHub]

- Built a bilingual (Arabic/English) spam detection system using NLTK and scikit-learn, applying advanced preprocessing, multiple vectorization methods (Count, N-grams, TF-IDF), and custom features, with a Random Forest model validated via cross-validation and feature importance analysis.

Customer Segmentation using Machine Learning [GitHub]

- analyzing an automotive dataset using KMeans clustering, Random Forest, and Linear Regression; performed feature engineering, outlier detection (DBSCAN), and dimensionality reduction (PCA) for actionable customer insights

Edge Speech-to-Text System with Whisper on Raspberry Pi [GitHub]

- Built a custom Yocto-based Embedded Linux distribution for Raspberry Pi 4 and deployed a containerized Whisper model for real-time audio transcription, using QEMU emulation for rapid iteration and OpenCV for computer vision support.

ECU embedded stack [GitHub]

- Developed a layered embedded software stack (MCAL, HAL, Service, Application) for AVR and ARM targets, validated through 4 test applications (keypad-LCD, interrupt-driven LED, I2C, ultrasonic distance sensing) with custom linker/startup code and CMake-based build automation.

Design, Modeling, and Control of a Six-DOF Artificial Upper Limb (Graduation Project) [GitHub]

- Modeled a 6-DOF robotic manipulator in MATLAB/Simulink, implementing forward and inverse kinematics to compute precise end-effector positioning and joint configurations.
- Developed path planning and trajectory generation algorithms, translating control laws into simulations to achieve smooth, stable, and accurate motion control.

LANGUAGES & ADDITIONAL SKILLS

Languages: Arabic (Native), English (Professional), German (Basic)

Leadership: Cross-functional Team Leadership, Agile/Scrum Methodologies, Technical Mentoring, Problem-Solving Under Constraints